Nombres complexes (partie 2)

Résumé de cours · Chapitre 9

I. Forme exponentielle

$e^{i\theta} = \cos\theta + i\sin\theta$; tout $z \neq 0$ s'écrit $z = re^{i\theta}$ ($r = |z|$, $\theta \equiv \arg z$).

$$e^{i\theta}e^{i\theta'} = e^{i(\theta+\theta')} \qquad \frac{e^{i\theta}}{e^{i\theta'}} = e^{i(\theta-\theta')} \qquad \overline{e^{i\theta}} = e^{-i\theta} \qquad (e^{i\theta})^n = e^{in\theta}$$

$e^{i\pi} = -1$, $e^{i\frac{\pi}{2}} = i$; $1 + e^{i\theta} = 2\cos\frac{\theta}{2}\, e^{i\frac{\theta}{2}}$.

II. Euler et Moivre

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \qquad (\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

Linéarisation : $\cos^2 x = \frac{1+\cos 2x}{2}$, $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^3 x = \frac{\cos 3x + 3\cos x}{4}$.

III. Équation $az^2 + bz + c = 0$ (a, b, c réels)

$$\Delta < 0 \ : \ z_{1,2} = \frac{-b \pm i\sqrt{-\Delta}}{2a} \qquad z_1 + z_2 = -\frac{b}{a} \qquad z_1 z_2 = \frac{c}{a}$$

IV. Transformations

Translation $\vec{w}(b)$	Homothétie (Ω, k)	Rotation (Ω, θ)
$z' = z + b$	$z' - \omega = k(z - \omega)$	$z' - \omega = e^{i\theta}(z - \omega)$

$z' = az + b$: $a = 1$ translation ; a réel $\neq 1$ homothétie ; $|a| = 1$, $a \neq 1$ rotation d'angle $\arg a$; centre $\omega = \frac{b}{1-a}$.